



DSA0405 - FUNDAMENTALS OF DATA SCIENCE

Data Manipulation

Cleaning, transforming, and shaping raw data for analysis

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What is Data Manipulation?

The process of adjusting, organizing, and reshaping raw data to make it easier to read, analyze, and use for decision-making.



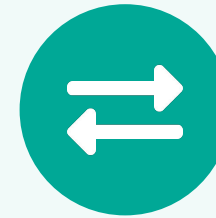
Turns messy, raw data into clean, structured tables



Prepares data for accurate analysis and visualization



A core step in every data science workflow



Raw Data



Manipulated Data

Clean · Structured · Ready to analyze

Core Data Manipulation Operations

The four building blocks used to reshape any dataset



Filter

Select rows that meet specific conditions



Sort

Arrange data in a meaningful order



Merge / Join

Combine data from multiple sources



Group & Aggregate

Summarize data by category

Handling Missing Data

Missing values are common in real datasets — here's how to deal with them



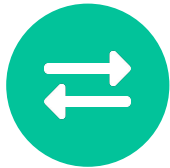
Detect

Identify missing or null values across columns



Remove

Drop rows or columns when too much data is missing



Impute

Fill gaps using the mean, median, or a fixed value

Tip: Always understand WHY data is missing before choosing how to handle it.

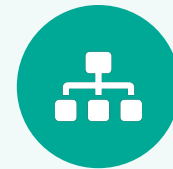
Data Transformation

Reshaping values so data is consistent and analysis-ready



Normalization

Rescale numeric values to a common range



Encoding

Convert categories into numeric form



Aggregation

Combine values into summary statistics



Reshaping

Pivot or melt data between wide and long formats

Popular Tools for Data Manipulation

The most common tools used across the data science workflow



Pandas

Python library for tabular data



SQL

Query and manipulate databases



Excel

Spreadsheet-based manipulation



Spark

Large-scale distributed data

Best Practices & Key Takeaways



Always keep a copy of the original raw data



Document every transformation step you apply



Check data types and ranges after each change



Validate results — don't assume the output is correct



Automate repeatable steps with scripts, not manual edits



Remember

Good data manipulation is invisible — it just makes every step after it easier.



Thank You

Fundamentals of Data Science · Data Manipulation